P R O D U C T D E S I G N C A S E S T U D Y

AccessPath

Inclusive city navigation. Find routes that work for your mobility needs, not just the shortest route.

Status. Portfolio concept. Prototype and design record completed 22 October 2026. Not validated with participants. Not a navigation service and not a claim of licensed sidewalk data.

Evidence base. Files in 03-NavAccessPath/, the PRD at 03-accesspath-inclusive-navigation.md, and the prototype in app/. Claims below are verified from those files, documented as hypotheses, or marked unknown.

1. Executive summary

AccessPath helps someone compare a few walking routes by the access settings they choose, and by how much of that comparison is actually known. The shortest walk is allowed to lose. The unique value proposition, taken from the PRD, is: find routes that work for your mobility needs, not just the shortest route.

The user problem is that a route which looks short can include stairs, a steep block, a missing curb ramp, or an entrance that has not been described. People often discover that after they have started. Some mainstream maps now offer a wheelchair or step-free option. The documented gap this concept addresses is that those options still tend to return one filtered route without showing, segment by segment, what was confirmed, what a person reported, and what nobody knows.

The proposed solution is explicit preferences, a comparison that keeps failing routes visible, and a narrow match label: “Matches your preferences on confirmed data.” Unknown stays unknown. A community report can raise a concern and cannot confirm a fact. Nothing in the interface is called fully accessible.

Implementation status. The must-have flow is working in a TypeScript browser prototype on illustrative Seattle sample data. Saved preferences exist only in this browser. There is no account, no live transit feed, no licensed basemap, and no turn-by-turn guidance. Nineteen automated tests passed on 22 October 2026. No participant study has been run.

Differentiator. AccessPath is a route comparison that shows evidence quality. It is not positioned as the first accessible map. The difference held against AccessMap, Google Maps, Citymapper, and Wheelmap is that conflicts stay listed, unknown blocks a match, and crowd notes never upgrade a gap to confirmed.

2. Product brief and objectives

Target users and jobs

The PRD names a primary audience of people who need step-free or otherwise accessible routes, and a secondary audience of caregivers, companions, and people temporarily navigating with luggage, injury, or a stroller. The product principle is that disabled people do not share one set of needs. Drivers, cyclists, and people who only want the fastest multimodal itinerary are not the first user to optimize.

The primary job, restated in 02-ux-research/jobs-to-be-done.md, is: when planning a trip, compare routes against access requirements so the choice has fewer unexpected barriers. In this MVP those requirements are avoid stairs, prefer curb ramps where known, and an optional maximum grade. The job is done when a route is chosen, its status matches the settings, and any unknown answer to those settings was visible. A barrier-free promise, turn-by-turn guidance, and a verified city work order are defined as not done.

Problem, goals, scope, and constraints

The problem statement in the PRD is a hypothesis: preference-based comparison and transparent confidence can make route planning more usable. It has not been tested with users. Product goals in the PRD are evaluation criteria, not measured results: 80 percent of prototype participants can configure a preference and select a suitable route; 80 percent can distinguish confirmed, community-reported, and unknown information; route-selection errors fall versus a documented baseline. The guardrail is never to label a route fully accessible when required data is unknown. The design goes further and never uses that phrase at all.

- Must have, and present on sample trips: preferences, comparison of distance, time, barriers, and confidence, explicit unknown, place and entrance detail, a barrier report, and a text alternative to the map.
- Should have, partly present: community reports with time and moderation status, a verified outage notice, and saved preferences. The notice is one sample. Saving is local, not an account.
- Could have later, not built: live transit integration, offline route notes, personal feedback history, and verified venue profiles.

- Non-goals: guaranteeing a barrier-free route, replacing emergency services or official guidance, and launching in many cities before data quality is reviewed.

Platform constraint: a mobile-first web prototype, responsive from a narrow phone to a wide browser. The pilot city is a hypothesis, Seattle, because published sidewalk-network work exists there. AccessPath has not licensed that data. Sample streets, grades, entrances, and reports are invented and labeled as such on every screen.

3. Product analysis and rationale

Positioning

01-product-analysis/uvp-and-positioning.md places AccessPath in route comparison for access conditions, rather than as a general map, a place-rating community, or a transit agency app. The positioning hypothesis: for people comparing a walking trip against stairs, curb ramps, and steepness, AccessPath shows whether each route matches those settings on confirmed data, where the data is only a community report, and where it is unknown. Unlike a single step-free toggle, it does not hide the route that fails, and it does not treat a gap as an all-clear. Reasons to believe are design decisions, not measured outcomes. Reasons someone might not switch, also hypotheses, include an existing trusted router, a need for live transit, a desire for a guarantee, or evidence language that feels slow when the only question was which way has no stairs.

Alternatives reviewed

The competitive scan in 01-product-analysis/competitive-analysis.md is limited to public pages reviewed on 22 October 2026. Absence of a feature is not asserted unless the source's described model makes the gap visible. Pricing is omitted except where a source states it. Citymapper's step-free routes are described by Citymapper as free.

Product	What the sources support	Open question
AccessMap (TCAT)	Personalized pedestrian routing: grade limits, raised curbs, stairs, street avoidance. Coverage discussed from Seattle toward OS-CONNECT. Public-transport mode recorded as no on the OSM wiki.	Whether route cards label unknown attributes or omit them. Confidence language was not fully audited.
Google Maps	Place attributes for wheelchair access in certain countries. A wheelchair-accessible route option. Help text defines an accessible entrance partly by width and by no steps or a ramp. Stair-free walking routes are reported by the New York Times (13 March 2026), which is journalism, not a help page.	Whether a walking result explains unknown blocks or only draws the filtered route.
Citymapper	Step-free trips that avoid stairs and wide gaps, with lift or ramp entrances and adjusted walk times. City by city. Issue reports are requested in the app.	Current route detail, grade limits, and what a report's status looks like after submission.
Wheelmap	A place map, not a router. Green, orange, red, and grey. Grey is unknown. Green means a specific step rule. The about page has described information on over 3.2 million places.	Out of job for route comparison. AccessPath does not claim to have invented unknown.

AXS Map and AccessNow appear in a July 2026 comparative paper and were not re-tested. Apple Maps wheelchair routing was seen only in secondary travel guides, not in Apple's help documentation. Ezymob was described in French press on 21 October 2026 and was not used. Those four are unknown for this analysis.

Decisions, priority, and trade-offs

Decision	Rationale	Trade-off
Four statuses, never "fully accessible."	PRD guardrail, extended so even a fully confirmed route is not given a blanket rating.	People who want a simple yes may find the label cautious. Untested.
Unknown blocks a match when that setting is on.	Silence should not read as step-free, level, or open.	Fewer routes can match. The no-match screen has to explain why.
Community notes cannot confirm.	A crowd all-clear is still a report.	Reports may feel weaker than users expect. Untested.
Conflicting reports stay listed.	Open PRD question, answered as a proposal. South arcade keeps two pending notes.	The screen refuses to pick a winner.
Preference fit is the default sort.	Leading with the shortest route would repeat the problem.	The list is longer to read. Shortest remains, with a warning to read status.
Presets do not name a condition.	Hypothesis that a "disability app" profile is the wrong object. Settings stay visible.	Three controls may feel like a form. Confidence in this assumption is low.
Schematic, not a basemap.	No license. The list carries every fact.	People may not recognize a street. Trust is untested.

Feature priority follows the PRD MoSCoW list, recorded in 03-information-architecture/feature-prioritization.md. Must items are what make the UVP visible, so they were built first. Live transit, offline notes, accounts, turn-by-turn, and multi-city launch were held. A blanket accessible score was explicitly not built.

Assumptions and risks

Nothing in 01-product-analysis/assumptions-and-risks.md has been confirmed with participants. Higher-risk assumptions: people want to see the failing route (medium confidence); “unknown” is preferable to a blank (medium); separate stair, curb, and grade settings are worth the controls (low); Seattle sidewalk data can support a pilot after licensing (low); refusing “fully accessible” increases trust (low). On-device preferences are judged high confidence for a test and low confidence for a product, because people may expect an account.

Harm from a wrong claim is addressed in the concept by the narrow match label, by the sentence that a match is not a guarantee, and by error and empty states that do not invent a path. Low-quality reports stay pending. Conflicting reports are not averaged. A pilot-area data-quality and safety review, required by the PRD before any public launch, has not been done.

4. UX research and evidence

Completed research. None with participants. 02-ux-research/research-findings.md states that no study has been run. The quote column is empty on purpose. The interview guide, screener, and research plan are instruments ready for a future study. They are not findings.

Methods actually used. Desk review of the PRD, public competitor pages, and author review of the prototype. Competitive sources and dates are in section 3. The author review is recorded in 07-prototype/prototype-review.md and is not a usability study.

Hypotheses the design is using

ID	Hypothesis	What would retire it
H1	People get partway down a short route before they learn about stairs.	Interviews where people already filter stairs and do not want the failing route shown.
H2	Silence about an attribute is misread as fine.	People who already treat blanks as unknown without help.
H3	One wheelchair mode does not cover a stroller day or a low-grade day.	People who say one step-free switch is the only control they will use.
H4	Crowd notes are useful only with a visible review state.	People who ignore crowd notes, or who trust them as much as a city survey.
H5	Conflicting notes should both remain visible.	People who need the product to pick one and will abandon a list.

Proto-personas and journey

02-ux-research/personas.md labels three proto-personas as hypotheses. Names are invented. They must not be quoted as users. Maya Chen is a hypothesis about someone who uses a power wheelchair and wants to point at what is confirmed before leaving. Andre Brooks is a hypothesis about a temporary cane user who does not want a diagnosis in the interface and who may delete settings later. Priya Shah is a hypothesis about different constraints on different days, including a stroller and a parent who uses a walker. Transit dispatchers, city asset managers, and professional access auditors are intentionally not personas yet.

02-ux-research/journey-map.md is a hypothetical current state, not an observed journey. Pain points in that map are assumptions: discovering a barrier after starting, treating a blank as safe, and having no review state on a crowd note. Forces in the jobs document are also hypotheses: a past failed trip as push, evidence as pull, the general map as habit, and anxiety that a new app could be confidently wrong.

Outstanding research questions

- Do people restate the match label without hearing a guarantee?
- Do they treat Pike and Seneca’s unknown grade as a reason not to call it a match?
- Do they use preference-fit sort, or switch to shortest and miss the conflict?
- Do preset names feel like diagnoses?
- Is pending read as already checked?
- Which city has data that can be licensed and is dense enough to test?

5. Information architecture and user flows

Primary navigation is Home, Plan, Preferences, and Reports. Comparison, route detail, entrances, and the report form sit under Plan. Saved preferences sit under Preferences. The tree is documented in 03-information-architecture/sitemap.md.

Primary journey

1. Home. Read the sample banner. The Seattle names are not surveyed measurements.
2. Preferences. Choose Step-free, or edit stairs, curb ramps, and a grade limit. Save.
3. Plan. Origin: Cal Anderson Park, south path. Destination: Pike Place Market, First Avenue entrance.
4. Compare. Read three statuses. Madison matches on confirmed data. Pike and Seneca has unknowns. Pine Street direct conflicts.
5. Inspect Madison gentle. Read segments, including the unknown surface on Western Avenue, which does not block the match because surface is not one of the settings.
6. Choose the route. Optionally open the south arcade entrance, where two pending reports are not merged, or report a barrier.
7. Confirmation says the report is pending review. It does not say the city has checked it. Withdraw remains available.

Alternate paths and recovery

Situation	Behavior
No confirmed match	Cal Anderson to Volunteer Park with step-free settings explains that no route matches. It does not draw a substitute and call it accessible.
Outside the sample	A destination such as Bainbridge is outside the pilot sample. No route is invented.
Service error	The comparison does not invent a path. The error state is a named prototype screen.
Offline	An offline state is shown. Offline route notes from the PRD’s could-have list are not implemented.
Location	A disclosure explains the sample start. The Geolocation API is not called. Declining permission stays on the trip screen.
Loading	A loading state exists so the comparison is not shown as empty data.
Shortest sort	Conflicts remain in the list. Copy warns the reader to check access status.
Changed preset	Gentle and stroller change outcomes on the sample, including the grade boundary: a grade equal to the limit is within the limit.

Interaction decisions that protect the task: Mark on the map sets a highlight and does not choose the route. The first route is marked initially. List and Map are tabs, and arrow keys move between them in the prototype. Back returns along the in-memory history. Refresh drops the in-memory trip and returns home. Preferences, reports, recent routes, and text scale persist only in this browser.

6. Wireframes and UI design

Screens

Fifteen annotated low-fidelity boards, S01 through S15, live in assets/wireframes/ as HTML and as 1240 by 760 PNG captures. Each board pairs a 360 by 680 phone frame with the user goal and numbered notes. S16 (your reports), S17 (outside the pilot), and S18 (location declined) are specified and, where noted, covered by prototype states rather than a separate wireframe PNG. The inventory is 04-wireframes/screen-inventory.md. High-fidelity UI is the browser prototype, specified in 06-high-fidelity-ui/. No Figma file was created.

Figure 1 is the comparison board. It is the decision the product is for: the longer route leads because it matches on confirmed data, the middle route is unknown, and the shortest route stays visible as a conflict.

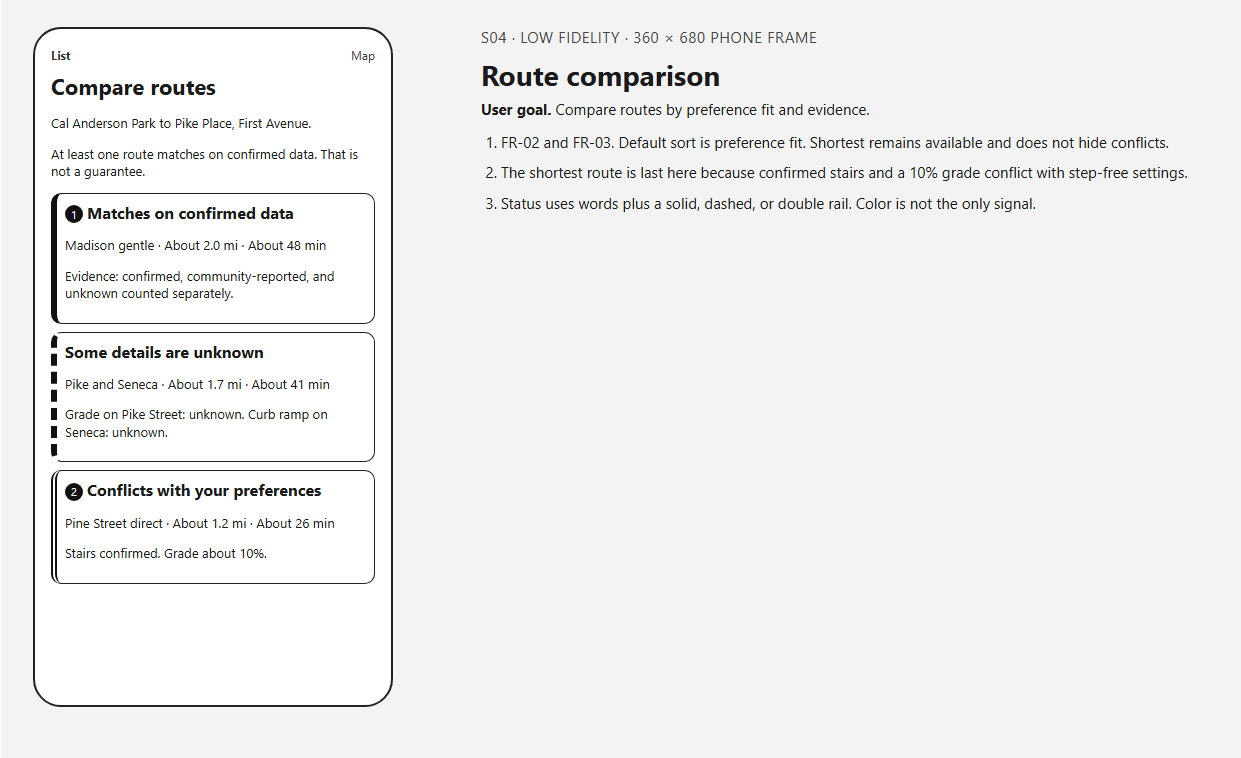


Figure 1. Low-fidelity route comparison, assets/wireframes/png/s04-results.png. Status uses words plus a solid, dashed, or double rail.

Layout, principles, and visual direction

Design principles in 05-ui-design-system/design-principles.md: evidence before reassurance; settings, not a diagnosis; the list is the product; a match is narrow; status is a word, with color, shape, and line pattern repeating it; reports stay humble; quiet chrome. The action color is a deep green. That green is not a message that a route is safe. The visual direction uses Atkinson Hyperlegible and Fraunces, with a paper ground (#F4F0E6) and ink (#14202B). Evidence is also encoded by shape so it is not color alone.

Reusable pieces, specified in 05-ui-design-system/components.md, include the sample banner, status chips, route cards, evidence counts, segment rows, the schematic and its legend, preset controls, report status, and dialogs. Stable vocabulary is in 06-high-fidelity-ui/content-and-microcopy.md. Buttons say what they do. “Mark on the map” replaced “Highlighted on the schematic” because list view hides the drawing on a narrow screen.

Responsive behavior and accessibility

Below 960 pixels the comparison stacks, and list mode hides the schematic. At 960 pixels and above, the schematic and the list can sit together. A text-size control sets the root font to 100, 125, or 150 percent. The prototype aims at WCAG 2.2 AA: keyboard access, visible focus, names on controls, a skip link, an aria-live region, and contrast pairs recorded in 05-ui-design-system/design-tokens.md. Calculated pairs include ink on paper at 14.52:1 and the corrected disabled button, #243140 on #E7E2D8, at 10.24:1. An earlier disabled pair was 3.88:1 and was replaced. These ratios were computed. They are not a formal audit. No NVDA, JAWS, or VoiceOver pass is documented, and 200 percent zoom was not measured in a session.

7. Implementation and prototype

The working prototype is app/. It is vanilla TypeScript with no UI framework. Source is compiled to app/dist and served by app/server.js, default http://127.0.0.1:4175. Domain rules live in app/src/evaluate.ts and are covered by app/test/evaluate.test.ts. Sample places and routes live in app/src/sample.ts. Setup is in README.md: npm install, npm test, npm start.

Implemented, simulated, and absent

Capability	State	Evidence
FR-01 Preferences change results, or the screen explains no match	Implemented on sample trips	evaluate tests; S02, S04, S11
FR-02 Distance, time, barriers, confidence	Implemented	Route cards and evidence counts
FR-03 Missing data stays unknown	Implemented	Unknown grade or curb blocks a match. A community all-clear does not confirm.

Capability	State	Evidence
FR-04 Segments and entrances in text	Implemented	Detail and place screens. The list is present in map view.
FR-05 Barrier report and status	Implemented locally	Submit creates Pending review. Withdraw works. Nothing is sent to a moderator.
FR-06 Signed-in save, change, delete	Partial	Change and delete work in localStorage. There is no account.
FR-07 Trusted outage feed	Simulated	One elevator notice labeled sample feed, not live.
Routing and map	Simulated	A few fixed origin-destination pairs and an SVG schematic. Not a router and not a licensed basemap.
Location	Disclosure only	No Geolocation API call.
Live transit, offline notes, multi-city, turn-by-turn	Not implemented	Held as later scope or non-goals.

Persistence keys are accesspath.preferences.v1, accesspath.reports.v1, accesspath.recent.v1, and accesspath.text-scale.v1. Review links of the form ?shot= hide the prototype controls and open home, preferences, trip, results, map, detail, place, report, status, saved, nomatch, outside, error, offline, location, and loading. shot=status and shot=saved write into that browser's storage. A sample verified report is labeled as not the viewer's.

Known prototype limits, from 07-prototype/prototype-flow.md: places are a fixed list; only some pairs have routes; reports never leave the browser; the results page is long on a phone; Madison's card can lead with an unknown surface even when the route matches, which is honest and easy to misread. The author review concludes that the prototype can demonstrate the UVP and the named failure states, and that it cannot support a claim that people understand it.

8. Testing and validation

Tests performed

On 22 October 2026, npm test in app/ compiled TypeScript and passed 19 node tests. They cover the step-free separation of match, unknown, and conflict; shortest sort still showing a conflict; unknown grade or curb blocking a match; a community all-clear not confirming; confirmed stairs and an over-limit grade conflicting; an unknown surface not blocking a match when surface is not selected; no step-free match to Volunteer Park; a grade equal to the stroller limit counting as within the limit; the gentle preset flagging a 7 percent grade without turning stairs on by itself; the sample elevator notice naming its source; status labels and copy omitting "fully accessible"; outside-pilot and missing pairs inventing no route; evidence counts; distance sums; reports starting pending and able to be withdrawn; HTML escaping; and the south arcade keeping two reports side by side.

Headless Chrome captured the 15 wireframe boards and 16 prototype review states under assets/. Those captures confirm that the screens render. They are not an interactive click-through and not an assistive-technology session. Contrast ratios in the design-token file were calculated with the WCAG relative-luminance formula.

Author changes, not usability findings

08-usability-testing/iteration-log.md separates author changes from future findings. No participant has used the prototype. Changes made while building, all dated 22 October 2026, include the ban on a blanket access label, the rule that a community all-clear cannot confirm, listing conflicting entrance reports without merging them, renaming the highlight control to "Mark on the map," raising disabled-button contrast from 3.88:1 to 10.24:1, and placing a sample banner on every screen. The findings template contains example rows that are explicitly marked as not findings. The five task scenarios in 08-usability-testing/test-scenarios.md are ready and unused.

Measures and what remains

No usability success rate, time-on-task, satisfaction score, or error reduction has been measured. The 80 percent PRD targets remain targets. A fair baseline for the selection-error metric has not been built, and this case study does not invent a delta. Outstanding validation is the interview plan, the five moderated tasks, an assistive-technology pass, a measured zoom pass, a check of current AccessMap, Google, and Citymapper route cards, and a pilot data-quality review before any public use.

9. Outcomes, limitations, and next steps

Verified deliverables. A complete documentation set from product analysis through a portfolio narrative, 15 annotated wireframe boards, an evidence-model diagram, a design system with computed contrast pairs, a

working TypeScript prototype of the core flow, 19 passing tests, and screenshot captures of the review states. Business impact and user improvement are not claimed, because no user study and no launch exist.

Strengths of the concept

- The UVP is visible without a narrator: the shortest sample route conflicts, and the reason is written as confirmed stairs and a confirmed grade.
- Match, unknown, and conflict are sentences, not a score.
- The same rules are encoded in tests, so a later copy change can be checked against the guardrail.
- Empty and error states refuse to invent a path.
- Research instruments exist, and they are labeled as unused.

Limitations and dependencies

- No participant evidence. Personas, journeys, and metric targets are hypotheses.
- Sample geography is invented. Following it on a street would be a misuse.
- FR-06 needs an account to meet the signed-in acceptance criterion. FR-07 needs a real feed with source and time.
- A production version depends on a licensed, maintained sidewalk network, a moderation process, and a safety review. None of those exist here.
- Accessibility is designed toward WCAG 2.2 AA and is not certified.

Highest-priority next work

1. Run the interview plan. Revise or delete personas that the notes do not support.
2. Run the five usability tasks and file notes with source references. Do not backfill quotes.
3. Retest the author changes against those sessions before treating them as settled.
4. Choose a pilot city and a data license only after that research.
5. Hold live routing, accounts, and transit alerts until a data-quality review exists.

10. Project artifact index

Paths are relative to 03-NavAccessPath/ unless noted. 00-project-brief/ does not exist. The source PRD is the sibling file 03-accesspath-inclusive-navigation.md.

Path	Contents
README.md	Product summary, folder guide, setup, review links, and remaining gaps.
01-product-analysis/	Brief, problem, cited competitive scan, positioning, assumptions, success metrics. Metrics are targets.
02-ux-research/	Plan, screener, interview guide, empty findings template, hypothesis personas, jobs, hypothetical journey.
03-information-architecture/	Sitemap, user flows, task analysis, MoSCoW and FR-01 to FR-07 traceability.
04-wireframes/	Inventory, specifications, annotations, edge cases. Visual boards are under assets/wireframes/.
05-ui-design-system/	Principles, visual direction, tokens with contrast pairs, components, accessibility notes.
06-high-fidelity-ui/	Screen specifications, microcopy, interaction specifications. The high-fidelity UI is the prototype.
07-prototype/	Flow, interaction states, and the author review. Application code is in app/, not in this folder.
08-usability-testing/	Test plan, five scenarios, empty findings template, author iteration log.
09-case-study/	Narrative, presentation outline, portfolio summary, this report, and overview-summary.md.
assets/diagrams/	evidence-model.svg. Three statuses: confirmed, community-reported, unknown.
assets/wireframes/	index.html, 15 HTML boards, wireframes.css, and png/s01 through s15.
assets/screenshots/	16 prototype PNGs, including results, map, detail, and the empty and error states.
app/	TypeScript prototype, tests, styles, and static server. Not product documentation.

11. Final status and evidence checklist

Deliverable	Evidence	Status	Remaining action
UVP and comparison concept	PRD; 01-product-analysis/; prototype results	Complete	Validate comprehension with participants.
FR-01 Preferences and no-match	app/src/evaluate.ts; S02, S04, S11	Complete	Sample trips only.
FR-02 Route comparison	Route cards; tests	Complete	None for the concept.
FR-03 Unknown left unknown	Tests for unknown grade, curb, and community all-clear	Complete	Check whether people still read unknown as safe.

Deliverable	Evidence	Status	Remaining action
FR-04 Text detail	Detail and entrance screens	Complete	None for the concept.
FR-05 Report status	Report flow; pending and withdraw tests	Partial	No moderator queue. Reports stay in the browser.
FR-06 Signed-in preferences	Saved screen; localStorage	Partial	No account.
FR-07 Verified alerts	One sample elevator notice	Partial	Not a feed.
Map and list, not color alone	S05, schematic legend, list	Partial	Schematic is not a basemap.
Wireframes	assets/wireframes/png/s01–s15	Complete	No Figma file. Documented.
Design system	05-ui-design-system/	Complete	Contrast is calculated, not audited.
Automated tests	07-prototype/prototype-review.md; 19 tests	Complete	Re-run after any rule change.
Participant research	02-ux-research/research-findings.md	Missing	Run the plan. Do not invent quotes.
Usability results	08-usability-testing/findings-template.md	Missing	Run the five scenarios.
WCAG 2.2 AA conformance	design-tokens.md; accessibility.md	Needs validation	Screen reader and zoom passes.
Pilot data and safety review	assumptions-and-risks.md	Needs validation	Seattle is a hypothesis. No license.
00-project-brief/	Not in the folder	Missing	PRD lives at the case-studies root. Not applicable as a folder.
Business or user outcome	No study, no launch	Missing	Do not claim impact.

Overall: the case study is implemented and documented. It is not user-validated. The safest reading of the work is that the product rules are explicit, testable, and honest about missing data, and that whether people can use those rules is still an open question.